

The existence and diversity of myxobacteria in lake mud – a previously unexplored myxobacteria habitat

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Summary

Myxobacteria are widely distributed in soil and oceanic sediment with a phylogeographic separation at high levels of classification. However, it is unclear whether freshwater environments, from which there has been no isolation report of myxobacteria since 1981, are habitats for myxobacteria. In this study, we investigated the presence of myxobacteria in lake mud using a two-step strategy. First, we constructed two universal bacterial libraries from the V3–V4 (V34) and V6–V8 (V678) hypervariable regions of 16S rRNA gene sequences. High-throughput 454 pyrosequencing revealed that myxobacteria were one of the major bacterial groups in the lake mud. They accounted for 5.77% of the total sequences and 7.52% of the total operational taxonomic units (OTUs) at a phylogenetic distance of 0.03. The community composition and taxonomic structure of the mud myxobacterial community were further analysed using myxobacteria-enriched libraries targeting the V34 and V678 regions, which were amplified with *Cystobacterineae*- and *Sorangineae*-specific primer pairs respectively. Phylogenetic analysis showed that the limnetic myxobacteria exhibited closer relationships to their soil than their marine relatives, but there were also exclusive taxa of limnetic myxobacteria detected. These results, together with a survey on available GenBank data, indicate that lake mud is a primary habitat for myxobacteria.

Introduction

Gram-negative myxobacteria, which are phylogenetically located in the δ -division of *Proteobacteria*, are famous for their cooperative behaviour (Shimkets, 1990; Dworkin, 1996) and production of secondary metabolites (Gerth

et al., 2003). Using fruiting body-dependent techniques, myxobacteria have been isolated from various soil-related samples, including soil, rotting wood, bark from dead or living trees and dung of herbivorous mammals, but rarely from aquatic environments (Reichenbach, 1999; Dawid, 2000). Accordingly, myxobacteria are widely recognized as typical soil bacteria (Reichenbach, 1999). Recently, halophilic (Iizuka *et al.*, 1998; 2003; Fudou *et al.*, 2002) and halotolerant (Li *et al.*, 2002) myxobacterial strains have been isolated from coastal areas. They have been shown to exhibit certain characteristics that differ from their soil-dwelling relatives (Zhang *et al.*, 2005; Wang *et al.*, 2007a). Molecular surveys have indicated that myxobacteria-related 16S rRNA gene sequences are also widely distributed in marine sediments at different depths and sites, but they are distantly related to those of soil myxobacteria (Jiang *et al.*, 2010; Brinkhoff *et al.*, 2012). The existence and distinct characteristics of marine myxobacteria not only suggest a lot of unexplored myxobacteria resources, but also provide insight into the geographic separation and environmental pressures on these microbes. However, it is still unknown whether freshwater environments are also native habitats for myxobacteria.

Several publications have reported the isolation of myxobacteria from river or lake (Brauss *et al.*, 1967; Gräf, 1975; Hook, 1977; Trzilová *et al.*, 1981). However, the accuracy of these early isolations was questioned because the limnetic isolates were highly similar in morphology to soil myxobacteria (Reichenbach, 1999). It has long been accepted that these myxobacterial isolates from aquatic environments germinated from myxospores or myxospores-containing fruiting bodies that had been washed or blown into rivers or lakes. To our knowledge, since 1981 (Trzilová *et al.*, 1981), there have been no reports of the isolation and identification of myxobacteria from aquatic environments, such as rivers or lakes. On the other hand, recent molecular surveys addressing aquatic bacterial communities based on traditional sequencing of 16S rRNA gene clone libraries from mud (Kalyuzhnaya *et al.*, 2008; Mueller-Spitz *et al.*, 2009; Ferrer *et al.*, 2011; Goffredi *et al.*, 2011) or high-throughput sequencing of metagenomic DNA from water samples (Clingenpeel *et al.*, 2011) have revealed scattered myxobacteria-related 16S rRNA gene sequences. It is therefore still uncertain whether myxobacteria are able to inhabit freshwater environments and whether the

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limnetic myxobacteria, if present, are phylogeographically separated from the myxobacteria living in other habitats. In this study, we investigated the presence of myxobacteria in lake mud and comparatively analysed the phylogeny of soil, marine and lake myxobacteria.

Results

Composition of the bacterial community in the lake mud

We had once isolated myxobacterial strains from mud of Chenghai Lake. Among the total of 113 isolates, nearly all of them belonged to the suborders of *Cystobacterineae* and *Sorangineae* (only one strain belong to *Nannocystis* in the suborder *Nannocystineae*), and had almost similar morphological characteristics to their soil relatives, even three strains belonged to neither of the known genera in *Cystobacterineae* (Supplementary materials S1). To evaluate the presence of myxobacteria in the lake mud, we analysed the bacterial communities using the V34 and V678 hypervariable regions of the 16S rRNA gene, which were amplified with two universal primer pairs U343F/U802R and U917F/U1407R (Table S1) respectively. The 454 pyrosequencing yielded 28 009 V34 and 38 287 V678 sequences, which are referred to as UV34 and UV678 libraries in this article. After pre-processing using Mothur v.1.20.2, 25 305 high-quality sequences for UV34 and 30 604 for UV678 were obtained, which is over the threshold of 25 000 tags for describing the composition of a bacterial community (Sogin, 2009). The data presented high coverage of the mud bacterial community. For example, at the 0.03 level (corresponding to the species level), the calculated number of OTUs for UV34 was 2546, and the Good's coverage (Good, 1953) was 0.94. Similarly, the OTU number and the Good's coverage for UV678 at the 0.03 level were 5559 and 0.87 (Table S2). These two libraries were combined for a more precise estimation of the composition and diversity of the bacterial community.

The dominant bacterial phyla in the combined library were *Proteobacteria*, *Actinobacteria*, *Acidobacteria* and *Chloroflexi*, accounting for 53.51%, 16.33%, 9.06% and 9.02% of the total number of sequences respectively (for details refer to Table S3). *Actinomycetales* of the *Actinobacteria* and *Pseudomonadales* of the γ -division of *Proteobacteria* were the most dominant orders, accounting for 7.41% and 6.73% of the total sequences in the libraries respectively (Fig. 1A). It was surprising and highly interesting to find that *Myxococcales* was the third largest bacterial order in the mud, accounting for 5.77% of the total sequences (7.25% in UV34 and 4.29% in UV678). Furthermore, at the 0.03 distance level, the OTU number for myxobacteria-related sequences reached 7.52% of the total (9.62% in UV34 and 5.41% in

UV678 respectively), which was the highest value in the mud (Table S4). The results suggested that *Myxococcales* was a dominant member of the bacterial community in the lake mud in terms of both population and species numbers.

Overall distribution pattern of Myxococcales in the lake mud

Three suborders have been described in *Myxococcales* thus far: *Cystobacterineae*, *Sorangineae* and *Nannocystineae*. These suborders are further classified into seven families, 20 genera and approximately 50 species (Garritty *et al.*, 2005; Shimkets *et al.*, 2006; Garcia *et al.*, 2009; 2010). However, at the 0.10 and 0.05 distance levels (corresponding to family and genus levels), the OTU numbers were 35 and 167 in UV34, 46 and 228 in UV678 respectively (Table S5). Among the myxobacteria-related sequences in the two universal libraries, *Cystobacterineae* was the largest suborder, accounting for 44.62% of the total sequences. The proportions of *Sorangineae* and *Nannocystineae* were 27.81% and 16.75% respectively. Including the unclassified sequences within the known suborders and families, the total unclassified sequences represented 50.14% and 36.99% of the myxobacteria-related sequences in the two universal libraries respectively (920/1835 in UV34 and 486/1314 in UV678). Of the cultured families, *Cystobacteraceae* was the most abundant myxobacterial family (constituting 32.71% of the total myxobacteria-related sequences), followed by *Polyangiaceae* (15.05%) and *Myxococcaceae* (9.15%). The predominant genus was *Archangium*, which is also one of the four most frequently encountered genera in soil (the other three cultured genera are *Polyangium*, *Myxococcus* and *Coralloccoccus*) (Dawid, 2000). Interestingly, *Anaeromyxobacter*, the only anaerobic genus described among *Myxococcales*, was the second most dominant myxobacterial genus, accounting for 9.14% of the total myxobacteria-related sequences in this mud sample. After *Archangium* and *Anaeromyxobacter*, *Kofleria* (8.77%), *Chondromyces* (4.45%) and *Phaselicystis* (4.15%) were among the abundant genera, whereas the other cultured genera each presented proportions below 3% (detailed information is given in Table S5). These results indicated that the unclassified *Myxococcales* in the lake mud comprise a large portion of the population and exhibit high species diversity.

Composition and taxonomic structure of Cystobacterineae and Sorangineae

To demonstrate the composition and taxonomic structure of the myxobacterial community in the mud more sufficiently, we further constructed two myxobacteria-enriched

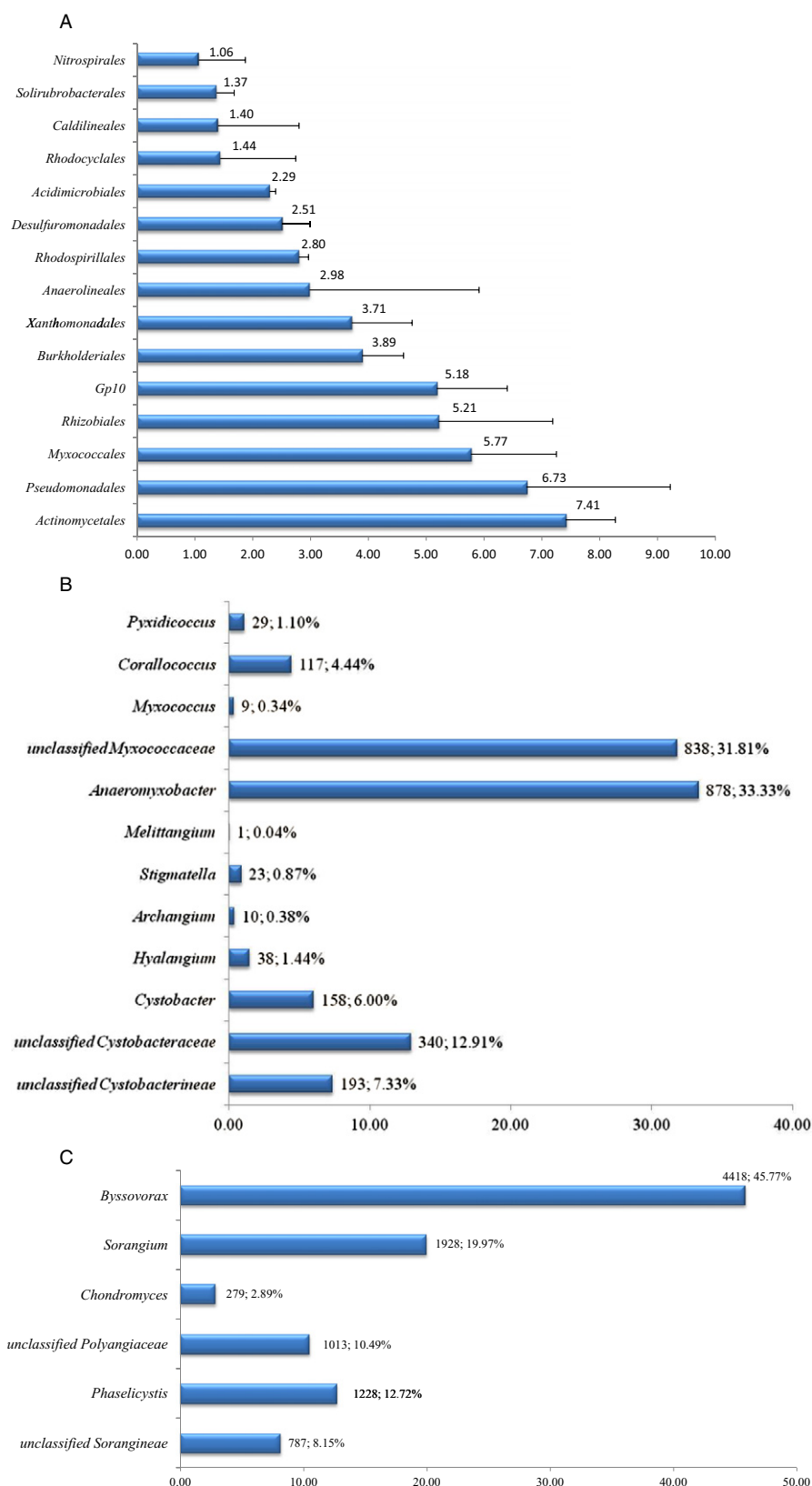


Fig. 1. The community structure of lake mud bacteria. The proportions of the predominant orders (A), measured using universal bacterial libraries. The composition of myxobacteria by genus: the number of distinct sequences present and the proportion (%) of corresponding genera in the suborder *Cystobacterineae* (B) and the suborder *Sorangineae* (C), measured using myxobacteria-enriched libraries.

Fig. 2. Comparative phylogenetic analyses of soil, marine and limnetic myxobacteria using the V3–V4 (A) and V6–V8 (B) regions of 16S rRNA gene sequences. Sequences with a similarity > 90% within the same sample type (soil, marine or limnetic environments) were removed from the analysis. The trees were constructed in a cycle using the MEGA 5.05 program. Bootstrap support was based on 1000 replicates. One unit along the bar is equivalent to one nucleotide change per 100 bp. In the phylogenetic trees, the numbers at the end of the names of the representatives of the lake mud myxobacterial sequences indicate the numbers of sequences they represent. The clades in the tree are separated by a phylogenetic distance of approximately 10%. The outermost cycle was added after construction.

libraries using the semi-specific primer pairs W1/802R and 917F/W6RC (Table S1). One of these libraries was for *Cystobacterineae*, targeting the V34 region (referred to as CV34), and the other was for *Sorangineae*, targeting the V678 region (referred to as SV678). After pre-processing using Mothur v.1.20.2, 9538 high-quality sequences were extracted from 11 664 CV34 reads. As expected, *Cystobacterineae* was the largest suborder in the CV34 library, accounting for 80.77% of the total myxobacteria-related sequences (2634 out of 3261). The proportions of *Sorangineae* and *Nannocystineae* were 12.63% and 4.57% respectively. In addition, there were 66 myxobacterial sequences, accounting for 2.02%, that could not be classified into the three known suborders. Similarly, 9788 high-quality SV678 sequences were extracted from 12 099 reads after pre-processing. A total of 9653 *Sorangineae* sequences, which accounted for 99.92% of the total myxobacterial-related sequences, were detected among the high-quality sequences.

The lake mud myxobacterial community included at least 32 families and 134 genera (based on the OTU numbers at the 0.10 and 0.05 distance levels) in the *Cystobacterineae* suborder (Table S2), of which only two families and nine genera have been identified. The proportions of the two identified families, *Cystobacteraceae* and *Myxococcaceae*, were 54.97% and 37.70%, respectively, while the left 7.33% belonged to unclassified *Cystobacterineae*. However, the total proportion of unclassified *Cystobacterineae* sequences (those that could not be clustered into any cultured *Cystobacterineae* genus) reached 52.05%. Consistent with that in the universal libraries, *Anaeromyxobacter* was among the most predominant genera, accounting for 33.33% of the total *Cystobacterineae* sequences. *Cystobacter* (6.00%) and *Corallococcus* (4.44%) were also among the dominant genera, while the other six cultured genera each presented a proportion of less than 2% (Fig. 1B; detailed information is given in Table S6).

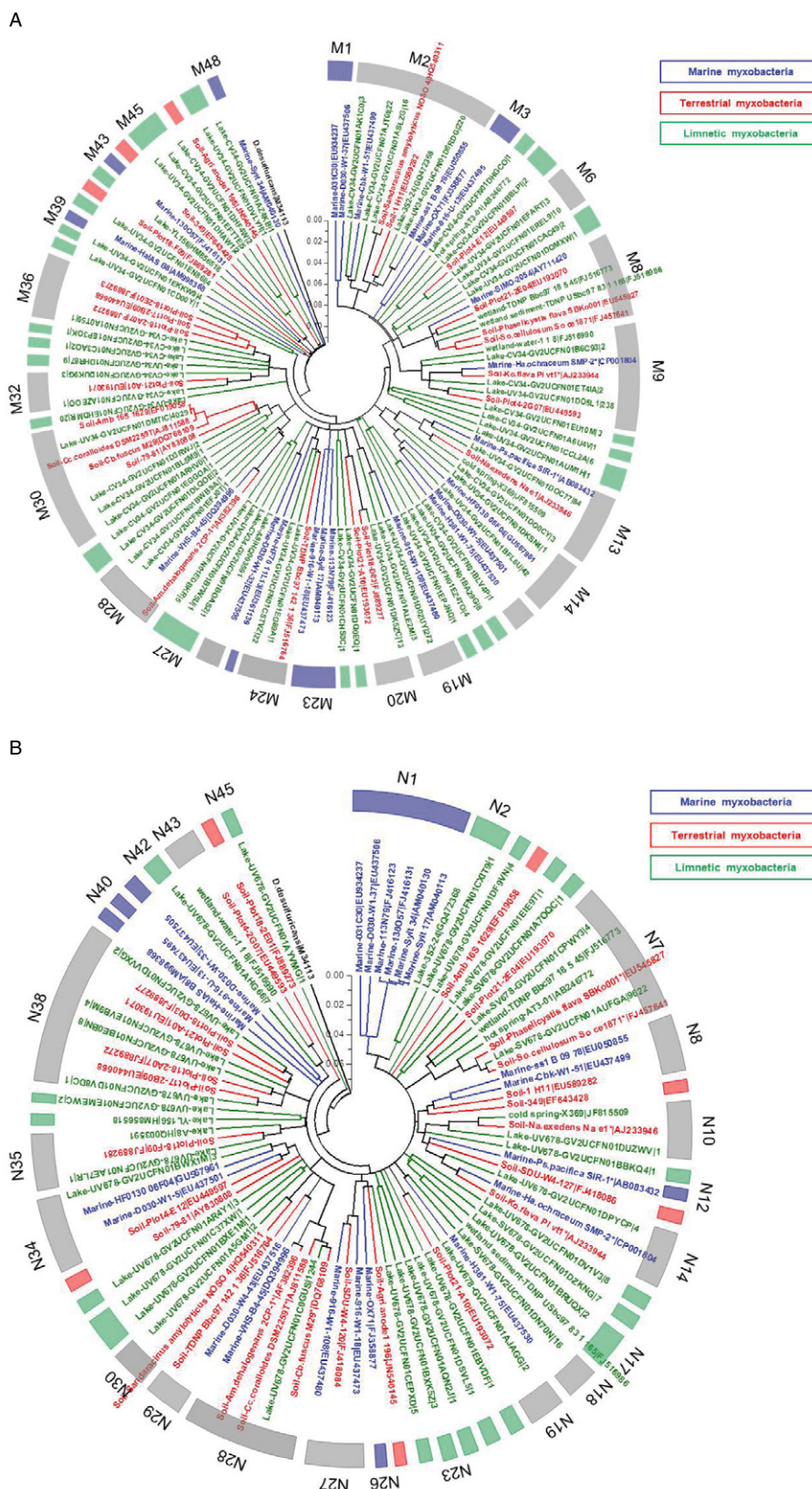
In the SV678 library, the number of OTUs at family level (the 0.10 distance), calculated from the 9653 SV678 *Sorangineae* sequences, was four, among which *Polyangiaceae* and the recently reported *Phaselicystidaceae* (Garcia *et al.*, 2009) have been identified. The proportions of *Polyangiaceae*, *Phaselicystidaceae* and unclassified *Sorangineae* were 79.13%, 12.72% and 8.15% respectively. The OTU number at genus level (the 0.05

distance) was 44, among which only four genera have been cultured (Fig. 1C). *Byssovorax* (Reichenbach *et al.*, 2006) contained the most of these sequences (4418; 45.77% of the total). The next most dominant groups were *Sorangium* (19.97%), *Phaselicystis* (12.72%) and *Chondromyces* (2.89%). The remaining 18.64% of sequences could not be classified into any cultured genus of the *Polyangiaceae* family or any cultured family of the *Sorangineae* suborder (detailed information is in Table S7).

Comparative phylogenetic analysis of soil, marine and lake myxobacteria

Because the terrestrial and marine myxobacteria are phylogeographically separated at high taxonomic levels (Jiang *et al.*, 2010; Brinkhoff *et al.*, 2012), there is a great deal of interest in analysing the phylogenetic relationships between the lake mud myxobacteria and their terrestrial and marine relatives. Hence, we constructed two comprehensive phylogenetic trees of myxobacteria using the sequences of the V34 (Fig. 2A) and V678 (Fig. 2B) regions of the 16S rRNA gene respectively. For simplification, representative sequences were selected at the 90% similarity level. The sequences were used for analysis if they were sufficiently long after alignment by Muscle version 3.8.31 (Robert, 2004) and if they were from typical soil, ocean or limnetic environments without apparent contamination. We chose myxobacteria-related sequences from the GenBank database using the search phrase 'Myxococcales 16S rRNA', as well as from correlational studies (Wu *et al.*, 2005; Jiang *et al.*, 2007; 2010; Brinkhoff *et al.*, 2012). The sequences used for tree construction had a minimum length of 372 bp, which is sufficient for relationship estimation at the 0.10 distance level.

The two phylogenetic trees constructed of myxobacterial sequences from different habitats were highly similar. At the 0.10 distance level, the sequences were grouped into roughly 48 (Fig. 2A) or 45 (Fig. 2B) clades. Consistent with previous findings (Jiang *et al.*, 2010; Brinkhoff *et al.*, 2012), the terrestrial and marine myxobacterial sequences were generally separated at high taxonomic levels. For example, the M1, M3, M23, M25, M39, M43 and M48 clades shown in Fig. 2A consisted of clusters of clannish marine myxobacteria (the corresponding clades



in Fig. 2B are N1, N26, N39 and N40). In addition to the clannish clusters, some marine myxobacterial sequences were also clustered with terrestrial sequences (M2, M8, M9 and M13 in Fig. 2A and N8, N14, N27, N29 and N34 in Fig. 2B). However, even within the same clades, there is normally a large phylogenetic distance between the marine and the terrestrial sequences. On the other hand, myxobacterial sequences from freshwater environments are often located on the same evolutionary branches with soil myxobacteria, forming terrestrial clades. For example, with the exception of a few individual marine sequences (such as Marine-VHS-B4-45IDQ394996 in M28 and N28), almost all sequences in the *Cystobacteriineae* suborder clades (M28, M29 and M30 in Fig. 2A, N28 in Fig. 2B) and *Sorangineae* suborder clades (M8 in Fig. 2A and N7 in Fig. 2B) came from soil and limnetic environments. However, in the valid *Nannocystineae* suborder (M9 and M13 in Fig. 2A and N10–N14 in Fig. 2B), marine, terrestrial and lake mud sequences were distributed miscellaneous. Interestingly, in addition to these shared clades, many lake mud myxobacterial sequences formed exclusive clades, such as M4, M5 and M7 in Fig. 2A, N2, N3 and N5 in Fig. 2B. The phylogenetic analysis clearly indicates that limnetic myxobacteria communities had form their special taxa in the long-term evolution.

Discussion

The molecular ecological studies described in this article indicated that, myxobacteria were among the dominant bacterial groups in the lake mud, not only of the abundance but also of the number of species. To testify the finding, we made a further survey on recently published pyrosequencing data. There are several papers concerning bacterial compositions in lake sediments (e.g. He *et al.*, 2012), in which, however, myxobacteria-related sequences were not mentioned by the authors. We analysed the retrieved sequence data from GenBank and found proportions of myxobacterial sequences changed significantly in lake sediments, and in some cases even accounted up to more than 10% of the total analysed sequences (Table S8). It is thus evident that there are numerous species of cultured and uncultured myxobacteria in freshwater environments.

Because limnetic myxobacterial sequences were often clustered in the same clades with soil myxobacterial sequences, the myxobacterial strains isolated from limnetic environments using classical myxobacterial isolation techniques were highly similar to their terrestrial relatives. However, comprehensive phylogenetic analyses of myxobacteria-related sequences from different habitats indicated that limnetic sequences not only shared clades with the sequences from other habitats,

but also formed exclusive limnetic clades. In contrast to the geographic separation of marine and soil environments, high similarity of soil and limnetic myxobacteria suggests that the soil myxobacteria are normally able to survive in river or lake mud, where they have adapted to water environments, and formed distinct water-based myxobacteria communities. For example, *Anaeromyxobacter*, the unique anaerobic genus described in *Myxococcales*, was first isolated from sediment taken from a small stream near Lansing, Mich (Cole *et al.*, 1994), and was later found to be widely distributed in various soils and sediments (Sanford *et al.*, 2002; 2007). *Anaeromyxobacter* strains are able to anaerobically grow by using many different terminal electron acceptors such as nitrate, fumarate and chlorophenolic compounds (Sanford *et al.*, 2002; He and Sanford, 2003; Marshall *et al.*, 2009; He and Yao, 2010). Because lake bottom sediment is a microaerobic environment, the *Anaeromyxobacter* group became predominant, which is obviously different from that in soil (Wu *et al.*, 2005; Jiang *et al.*, 2007) or marine environments (Jiang *et al.*, 2010; Brinkhoff *et al.*, 2012). Probably similar to the adaptation of marine myxobacteria to the oceanic environments (Zhang *et al.*, 2005; Wang *et al.*, 2007a), these limnetic organisms may have developed a set of distinct characteristics that distinguish from their soil or marine relatives.

Experimental procedures

Sample collection

Mud samples were collected from Chenghai Lake (26°27'–26°38'N, 100°38'–100°41'E), which is located in the Yunnan province of China. The annual average temperature of the lake water is approximately 16°C, with a maximum of 31°C in August and a minimum of 2°C in January (Tao *et al.*, 1999; Wan *et al.*, 2005). Chenghai Lake is a moderately nutrient-enriched alkaline lake with an average pH of 8.5 and total ion concentration of 961 mg l⁻¹ (Tao *et al.*, 1999; Wan *et al.*, 2005). The sediments of Chenghai Lake have an average organic C, H, N of 1.0%, 0.7% and 0.2% respectively (Wan *et al.*, 2005). Mud was sampled at a distance of 50 m from the shore in August 2008. After collection, the samples were air dried on a super-clean bench and then stored at –20°C.

DNA extraction

Total DNA was extracted from 1.0 g of mud using the FastDNA® SPIN Kit for Soil (MP Biomedicals, USA) following the manufacturer's instructions. The extracted DNA was then stored at –20°C. For sequencing, a DNA sample was dissolved in sterilized double distilled water (ddH₂O), and was used within the next a few days. The concentration of the DNA solution was determined using a NanoDrop® spectrophotometer ND-1000 (USA).

Construction of 16S rRNA gene sequence libraries and sequencing

The primer pairs U343F/U802R and U917F/U1407R, modified from Nossa *et al.* (2010), are universal primers for the bacterial 16S rRNA gene. They target the V3–V4 (referred to as V34 in this article) and V6–V8 (referred to as V678) regions respectively. The semi-specific primer pair W1/802R was designed to selectively amplify the V34 fragments of *Cystobacterineae* 16S rRNA gene sequences. The semi-specific 917F/W678 primer pair was designed to target the V678 region of *Sorangineae*. The forward primer used in 454 pyrosequencing contained the 454 Life Sciences primer A sequence, a four-base linker sequence (the key sequence, GACT), a unique barcode (the tag sequence for separate PCR products) and the corresponding forward primer for the 16S rRNA genes. The reverse primer contained the 454 Life Sciences primer B sequence, the four-base linker sequence (GACT) and the corresponding reverse primer for the 16S rRNA gene. Detailed information on the primers is shown in Table S1.

To minimize the amplification bias, the techniques with a lower annealing temperature, shorter extension time (< 180 s) and fewer cycles were employed in the PCR procedures (Ishii and Fukui, 2001; Kanagawa, 2003; Kurata *et al.*, 2004; Acinas *et al.*, 2005). PCR amplifications were performed using the following procedure. A 20 µl reaction mixture contained 0.5 µM forward and reverse primers, 50 ng of template DNA, 10 µl of 2× GC buffer II (TAKARA, Japan), 0.2 mM dNTPs, certified DNA-free PCR water and 0.5 U of LA Taq (TAKARA, Japan). The thermal cycling programme consisted of an initial denaturation at 94°C for 3 min, followed by 25 cycles of denaturation at 94°C for 30 s, annealing at 57°C for 30 s and extension at 72°C for 30 s, with a final extension of 10 min at 72°C. The replicate amplification products were pooled together. After detection in 1.0% agarose gels (Invitrogen, USA), the products were purified using the TIANGel Midi Purification Kit (TIANGEN, China) according to the manufacturer's instructions. The final concentration of the pooled DNA was determined using a NanoDrop® spectrophotometer ND-1000 (USA). Pyrosequencing was carried out on a 454 Life Sciences Genome Sequencer FLX (Roche) at the Shanghai Majorbio Bio-Pharm Biotechnology, China.

Sequence pre-processing and analysis

The 454 Life Sciences primer A sequence, Key sequence and Tag sequence were removed from the sequencing reads. Mothur v.1.20.2 software (Schloss *et al.*, 2009) was employed to remove sequences that were shorter than 200 bp, contained any ambiguous bases, exhibited a homopolymer of longer than 8 bp, or were chimera. After pre-processing, the obtained high-quality sequences were used in the following bioinformatics analyses.

The operational taxonomic units (OTUs) and diversity parameters of the UV34 and UV678 universal bacterial libraries were generated via the nearest neighbour cluster method using Mothur v.1.20.2 (Schloss *et al.*, 2009). Then, the sequences were submitted to the online Ribosomal Database Project (RDP) classifier to determine their taxonomy

(<http://rdp.cme.msu.edu/>) (Wang *et al.*, 2007b), setting the bootstrap cut-off at 50%. Myxobacteria-related sequences were selected and phylogenetically analysed to describe overall distribution pattern. For the *Cystobacterineae*- and *Sorangineae*-specific libraries, CV34 and SV678, the OTUs and diversity parameters were generated from the *Cystobacterineae* and *Sorangineae* sequences which were selected by the RDP classifier. Representative sequences of myxobacteria were selected from the UV34/CV34 and UV678/SV678 libraries for the construction of phylogenetic trees with reference sequences of myxobacteria-related sequences from soil, marine and other limnetic environments to compare their phylogenetic relationships. Sequences were aligned with MUSCLE (Robert, 2004) version 3.8.31, and after alignment, phylogenetic trees were built using the neighbour joining program in MEGA5.05 (Tamura *et al.*, 2011).

GenBank accession numbers

The sequence data have been submitted to the GenBank database under the Accession No. SRP011411.5.

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References

- Acinas, S.G., Sarma-Rupavtarm, R., Klepac-Ceraj, V., and Polz, M.F. (2005) PCR-induced sequence artifacts and bias: insights from comparison of two 16S rRNA clone libraries constructed from the same sample. *Appl Environ Microbiol* **71**: 8966–8969.
- Brauss, F.W., Heyne-Katzenberger, I., Pech, H., and Barth, H. (1967) Beiträge zur Mikrobiologie von Binnengewässern (I). *Arch Hyg Bakteriol* **150**: 717–724.
- Brinkhoff, T., Fischer, D., Vollmers, J., Voget, S., Beardsley, C., Thole, S., *et al.* (2012) Biogeography and phylogenetic diversity of a cluster of exclusively marine myxobacteria. *ISME J* **6**: 1260–1272.
- Clingenpeel, S., Macur, R.E., Kan, J.J., Inskeep, W.P., Lovalvo, D., Varley, J., *et al.* (2011) Yellowstone Lake: high-energy geochemistry and rich bacterial diversity. *Environ Microbiol* **13**: 2172–2185.
- Cole, J.R., Cascarelli, A.L., Mohn, W.W., and Tiedje, J.M. (1994) Isolation and characterization of a novel bacterium growing via reductive dehalogenation of 2-chlorophenol. *Appl Environ Microbiol* **60**: 3536–3542.
- Dawid, W. (2000) Biology and global distribution of myxobacteria in soils. *FEMS Microbiol Rev* **24**: 403–427.
- Dworkin, M. (1996) Recent advances in the social and developmental biology of the myxobacteria. *Microbiol Mol Biol Rev* **60**: 70–102.
- Ferrer, M., Guazzaroni, M.E., Richter, M., García-Salamanca, A., Yarza, P., Suárez-Suárez, A., *et al.* (2011) Taxonomic and functional metagenomic profiling of the microbial community in the anoxic sediment of a sub-saline

- shallow lake (Laguna de Carrizo, Central Spain). *Microb Ecol* **62**: 824–837.
- Fudou, R., Jojima, Y., Iizuka, T., and Yamanaka, S. (2002) *Haliangium ochraceum* gen. nov., sp. nov. and *Haliangium tepidum* sp. nov.: novel moderately halophilic myxobacteria isolated from coastal saline environments. *J Gen Appl Microbiol* **48**: 109–116.
- Garcia, R., Gerth, K., Stadler, M., Dogma, I.J., and Müller, R. (2010) Expanded phylogeny of myxobacteria and evidence for cultivation of the 'unculturable'. *Mol Phylogenet Evol* **57**: 878–887.
- Garcia, R.O., Reichenbach, H., Ring, M.W., and Müller, R. (2009) *Phaselicystis flava* gen. nov., sp. nov., an arachidonic acid-containing soil myxobacterium, and the description of *Phaselicystidaceae* fam. nov. *Int J Syst Evol Microbiol* **59**: 1524–1530.
- Garrity, G.M., Brenner, D.J., and Krieg, N.R. (2005) *Bergey's Manual of Systematic Bacteriology*, 2nd edn, Vol. 2. New York, USA: Springer-Verlag.
- Gerth, K., Pradella, S., Perlova, O., Beyer, S., and Müller, R. (2003) Myxobacteria: proficient producers of novel natural products with various biological activities – past and future biotechnological aspects with the focus on the genus *Sorangium*. *J Biotechnol* **106**: 233–253.
- Goffredi, S.K., Kantor, A.H., and Woodside, W.T. (2011) Aquatic microbial habitats within a neotropical rainforest: bromeliads and pH-associated trends in bacterial diversity and composition. *Microb Ecol* **61**: 529–542.
- Good, I.J. (1953) The population frequencies of species and the estimation of population parameters. *Biometrika* **40**: 237–264.
- Gräf, W. (1975) Myxobakterien der Gattung *Myxococcus* als indirekte Fäkalstoffindikatoren in Oberflächen-gewässern. *Zbl Bakt Hyg I Abt Orig B* **160**: 28–39.
- He, Q., and Sanford, R.A. (2003) Characterization of Fe(III) reduction by chlororespiring *Anaeromyxobacter dehalogenans*. *Appl Environ Microbiol* **69**: 2712–2718.
- He, Q., and Yao, K. (2010) Microbial reduction of selenium oxyanions by *Anaeromyxobacter dehalogenans*. *Bioresour Technol* **101**: 3760–3764.
- He, R., Wooller, M.J., Pohlman, J.W., Catranis, C., Quensen, J., Tiedje, J.M., and Leigh, M.B. (2012) Identification of functionally active aerobic methanotrophs in sediments from an arctic lake using stable isotope probing. *Environ Microbiol* **14**: 1403–1419.
- Hook, L.A. (1977) Distribution of myxobacters in aquatic habitats of an alkaline bog. *Appl Environ Microbiol* **34**: 333–335.
- Iizuka, T., Jojima, Y., Fudou, R., and Yamanaka, S. (1998) Isolation of myxobacteria from the marine environment. *FEMS Microbiol Lett* **169**: 317–322.
- Iizuka, T., Jojima, Y., Fudou, R., Tokura, M., Hiraishi, A., and Yamanaka, S. (2003) *Enhygromyxa salina* gen. nov., sp. nov., a slightly halophilic myxobacterium isolated from the coastal areas of Japan. *Syst Appl Microbiol* **26**: 189–196.
- Ishii, K., and Fukui, M. (2001) Optimization of annealing temperature to reduce bias caused by a primer mismatch in multitemplate PCR. *Appl Environ Microbiol* **67**: 3753–3755.
- Jiang, D.M., Wu, Z.H., Zhao, J.Y., and Li, Y.Z. (2007) Fruiting and non-fruited myxobacteria: a phylogenetic perspective of cultured and uncultured members of this group. *Mol Phylogenet Evol* **44**: 545–552.
- Jiang, D.M., Kato, C., Zhou, X.W., Wu, Z.H., Sato, T., and Li, Y.Z. (2010) Phylogeographic separation of marine and soil myxobacteria at high levels of classification. *ISME J* **4**: 1520–1530.
- Kalyuzhnaya, M.G., Lidstrom, M.E., and Chistoserdova, L. (2008) Real-time detection of actively metabolizing microbes by redox sensing as applied to methylotroph populations in Lake Washington. *ISME J* **2**: 696–706.
- Kanagawa, T. (2003) Bias and artifacts in multitemplate polymerase chain reactions (PCR). *J Biosci Bioeng* **96**: 317–323.
- Kurata, S., Kanagawa, T., Magariyama, Y., Takatsu, K., Yamada, K., Yokomaku, T., and Kamagata, Y. (2004) Reevaluation and reduction of a PCR bias caused by reannealing of templates. *Appl Environ Microbiol* **70**: 7545–7549.
- Li, Y.Z., Hu, W., Zhang, Y.Q., Qiu, Z., Zhang, Y., and Wu, B.H. (2002) A simple method to isolate salt-tolerant myxobacteria from marine samples. *J Microbiol Methods* **50**: 205–209.
- Marshall, M.J., Dohnalkova, A.C., Kennedy, D.W., Plymale, A.E., Thomas, S.H., Löffler, F.E., et al. (2009) Electron donor-dependent radionuclide reduction and nanoparticle formation by *Anaeromyxobacter dehalogenans* strain 2CP-C. *Environ Microbiol* **11**: 534–543.
- Mueller-Spitz, S.R., Goetz, G.W., and McLellan, S.L. (2009) Temporal and spatial variability in nearshore bacterioplankton communities of Lake Michigan. *FEMS Microbiol Ecol* **67**: 511–522.
- Nossa, C.W., Oberdorf, W.E., Yang, L.Y., Aas, J.A., Paster, B.J., DeSantis, T.Z., et al. (2010) Design of 16S rRNA gene primers for 454 pyrosequencing of the human foregut microbiome. *World J Gastroenterol* **16**: 4135–4144.
- Reichenbach, H. (1999) The ecology of the myxobacteria. *Environ Microbiol* **1**: 15–21.
- Reichenbach, H., Lang, E., Schumann, P., and Sproer, C. (2006) *Byssovorax cruenta* gen. nov., sp. nov., nom. rev., a cellulose-degrading myxobacterium: rediscovery of 'Myxococcus cruentus' Thaxter 1897. *Int J Syst Evol Microbiol* **56**: 2357–2363.
- Robert, E.L. (2004) MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res* **32**: 1792–1797.
- Sanford, R.A., Cole, J.R., and Tiedje, J.M. (2002) Characterization and description of *Anaeromyxobacter dehalogenans* gen. nov., sp. nov., an aryl-halo-respiring facultative anaerobic myxobacterium. *Appl Environ Microbiol* **68**: 893–900.
- Sanford, R.A., Wu, Q., Sung, Y., Thomas, S.H., Amos, B.K., Prince, E.K., and Löffler, F.E. (2007) Hexavalent uranium supports growth of *Anaeromyxobacter dehalogenans* and *Geobacter* spp. with lower than predicted biomass yields. *Environ Microbiol* **9**: 2885–2893.
- Schloss, P.D., Westcott, S.L., Ryabin, T., Hall, J.R., Hartmann, M., Hollister, E.B., et al. (2009) Introducing mothur: open-source, platform-independent, community-supported software for describing and comparing microbial communities. *Appl Environ Microbiol* **75**: 7537–7541.
- Shimkets, L., Dworkin, M., and Reichenbach, H. (2006) The myxobacteria. In *The Prokaryotes*, 3rd edn, Vol. 7. Dworkin, M., Falkow, S., Rosenberg, E., Schleifer, K.-H.,

- and Stackebrandt, E. (eds). Heidelberg: Springer, pp. 31–115.
- Shimkets, L.J. (1990) Social and developmental biology of the myxobacteria. *Microbiol Mol Biol Rev* **54**: 473–501.
- Sogin, M.L. (2009) Characterizing microbial population structures through massively parallel sequencing. In *Uncultivated Microorganisms*. Epstein, S.S. (ed.). Heidelberg: Springer-Verlag, pp. 19–34.
- Tamura, K., Peterson, D., Peterson, N., Stecher, G., Nei, M., and Kumar, S. (2011) MEGA5: molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. *Mol Biol Evol* **28**: 2731–2739.
- Tao, W., Xia, F., and Xing, C. (1999) On environmental issues of Lake Chenghai and its management strategy. *Resour Environ Yangtze Basin* **8**: 210–214 (in Chinese).
- Trzilová, B., Miklošovicová, L., Morhácová, G., and Golaisová, E. (1981) Die Wasserqualität der Zuflüsse des tschechoslowakischen Donauabschnittes von limnologischer und hygienischer Sicht. *Biológia* **36**: 765–774.
- Wan, G., Chen, J., Wu, F., Xu, S., Bai, Z., Wan, E., *et al.* (2005) Coupling between $^{210}\text{Pb}_{\text{ex}}$ and organic matter in sediments of a nutrient-enriched lake: an example from Lake Chenghai, China. *Chem Geol* **224**: 223–236 (in Chinese).
- Wang, B., Hu, W., Liu, H., Zhang, C.Y., Zhao, J.Y., Jiang, D.M., *et al.* (2007a) Adaptation of salt-tolerant *Myxococcus* strains and their motility systems to the ocean conditions. *Microb Ecol* **54**: 43–51.
- Wang, Q., Garrity, G.M., Tiedje, J.M., and Cole, J.R. (2007b) Naive Bayesian classifier for rapid assignment of rRNA sequences into the new bacterial taxonomy. *Appl Environ Microbiol* **73**: 5261–5267.
- Wu, Z.H., Jiang, D.M., Li, P., and Li, Y.Z. (2005) Exploring the diversity of myxobacteria in a soil niche by myxobacteria-specific primers and probes. *Environ Microbiol* **7**: 1602–1610.
- Zhang, Y.Q., Li, Y.Z., Wang, B., Wu, Z.H., Zhang, C.Y., Gong, X., *et al.* (2005) Characteristics and living patterns of marine myxobacterial isolates. *Appl Environ Microbiol* **71**: 3331–3336.

Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. The primers used for the construction of 16S rRNA gene libraries and 454 pyrosequencing.

Table S2. Detailed diversity parameters of the four libraries.

Table S3. The bacterial community composition of the lake mud at the levels of phylum and order, determined by the libraries of UV34 and UV678.

Table S4. The OTU numbers of bacteria in phylum and order level at 3% phylogenetic distance.

Table S5. The composition of myxobacteria, calculated by bacteria universal libraries UV34 and UV678.

Table S6. The composition of suborder *Cystobacterineae*, calculated by *Cystobacterineae* enriched libraries CV34.

Table S7. The composition of suborder *Sorangineae*, calculated by *Sorangineae* enriched libraries SV678.

Table S8. Surveys of the appearance of myxobacteria-related sequences in the pyrosequencing data of lake sediments (released by He *et al.*, 2012; retrieved from GenBank).

Supplementary materials S1. Isolation of myxobacteria strains from mud samples of Chenghai Lake.

Fig. S1-1. Colony photographs of *Myxococcus* (A) and *Sorangium* (B) strains isolated from Chenghai Lake mud. Bar in (A) is equal to 1.0 mm for 0558-ZXW144 and 0.5 mm for others. Bar in (B) is equal to 1.0 cm for So0558-31 and 0.5 cm for others.

Fig. S1-2. Phylogenetic analysis of the isolated myxobacteria stains based on 16S rRNA gene sequences. The bar is equivalent to two nucleotides changes per 100 bp. The numbers on branch nodes indicate bootstrap support percentages based on 1000 replications. The sequences from Chenghai Lake sediment are underlined for easy tracking.